

CASE REPORT OPEN ACCESS

Lung Adenocarcinoma Diagnosed in a Nodule After Initial Spontaneous Regression in Size

Lin Mo^{1,2}  | Filipe Vilas Boas^{1,3} | Shileigh Martin¹ | Jessica Rajanayagam¹ | Vicki Pretorius¹

¹Top End Health Service, Royal Darwin Hospital, Darwin, Australia | ²College of Medicine and Public Health, Flinders University, Adelaide, Australia | ³Radiology SA, Adelaide, Australia

Correspondence: Lin Mo (linmo388@gmail.com)

Received: 28 August 2025 | **Revised:** 26 September 2025 | **Accepted:** 7 October 2025

Associate Editor: James C. M. Ho

Funding: The authors received no specific funding for this work.

Keywords: lung cancer | lung nodule | screening | spontaneous regression

ABSTRACT

Widespread adoption of lung cancer screening programmes increases the detection of pulmonary nodules, both benign and malignant. Although guidelines exist to direct the management of nodules, they do not address the management of nodules that display atypical behaviour. Although rare, certain histotypes of cancer demonstrate spontaneous resolution without treatment. In some cases, this heralds a long-lasting remission, whereas in others the cancer may progress after initial regression. We describe partial spontaneous regression of a pulmonary nodule that subsequently progressed and was diagnosed as adenocarcinoma.

1 | Introduction

Lung cancer remains the leading cause of cancer-related mortality globally [1]. Lung cancer screening trials, including the National Lung Cancer Screening Trial (NLST) and the NELSON trial, have demonstrated that lung cancer screening with low-dose computed tomography (CT) is effective at reducing lung cancer mortality [2, 3]. However, these screening programmes detect many lung nodules, the vast majority of which are benign [4]. Various strategies and guidelines have been implemented for lung nodule management, including Fleischner Society Guidelines, British Thoracic Society Guidelines and LUNG-RADS guidelines for screen-detected nodules [5–7]. These provide guidance on the management of stable or enlarged nodules but do not address nodules that have partially resolved. We present a case of lung adenocarcinoma spontaneously regressing after initial CT before progressing again 1 year later.

2 | Case Report

A 78-year-old man was referred to the respiratory clinic for ongoing management of his severe chronic obstructive pulmonary disease (COPD) with a 60-pack-year smoking history, having quit smoking several years ago. His past medical history included a treated prostatic abscess and gastroesophageal reflux disease. He did not have any personal or family history of cancer. Clinical examination was unremarkable, and the patient reported mild shortness of breath on exertion. Spirometry demonstrated a forced expiratory volume in 1 s (FEV1) of 1.06 L or 36% of predicted and a diffusing capacity, corrected for haemoglobin, of 61% predicted.

In September 2021, he was admitted to the High Dependency Unit of the hospital with community-acquired pneumonia. A CT scan of his chest during this admission demonstrated extensive

This is an open access article under the terms of the [Creative Commons Attribution-NonCommercial](https://creativecommons.org/licenses/by-nc/4.0/) License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited and is not used for commercial purposes.

© 2025 The Author(s). *Respirology Case Reports* published by John Wiley & Sons Australia, Ltd on behalf of The Asian Pacific Society of Respirology.

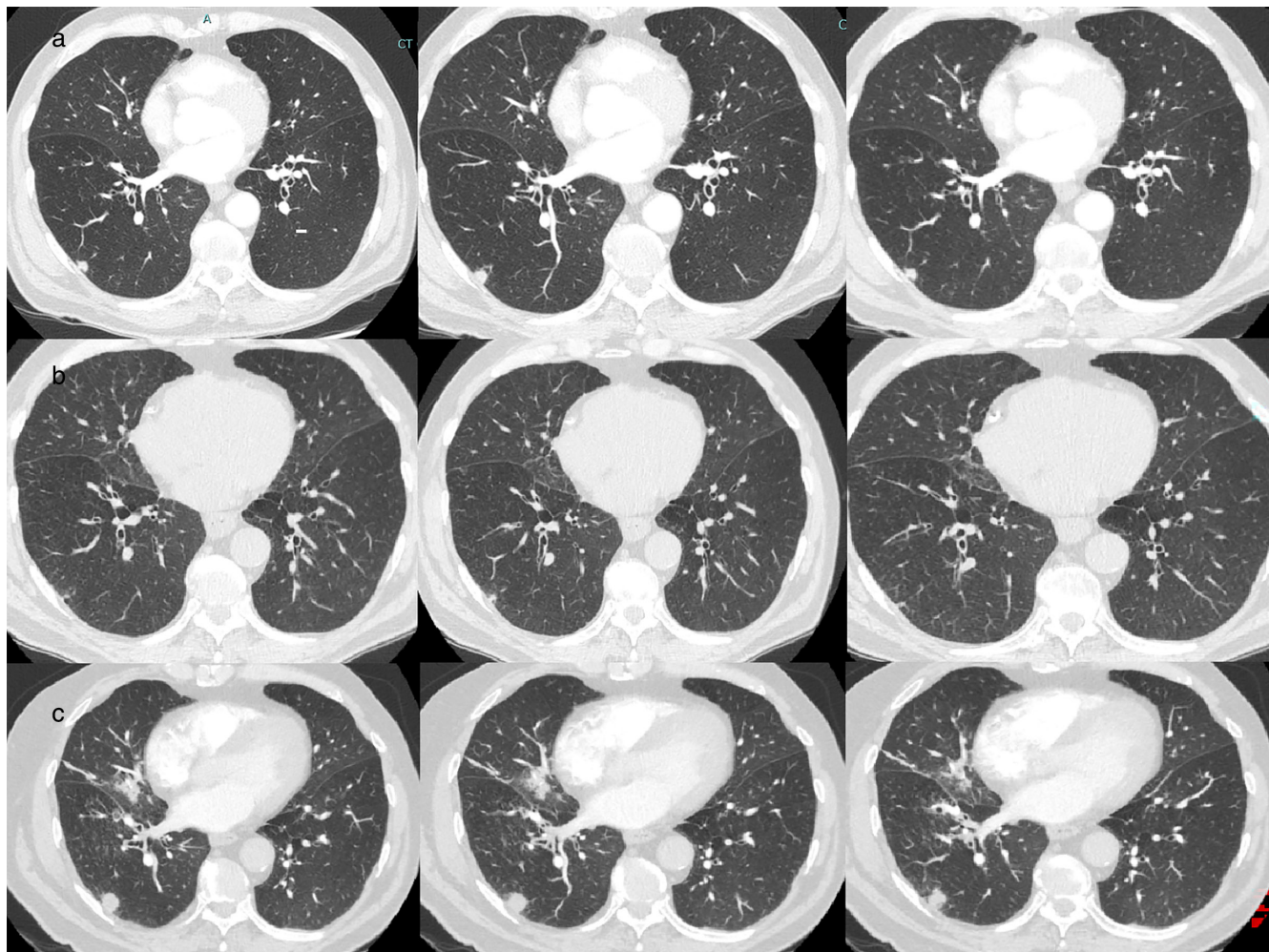


FIGURE 1 | (a) (March 2022): 1 mm Axial CT with three contiguous slices taken two slices apart demonstrating an 8-mm right lower lobe subpleural solid nodule. (b) (December 2022) 1 mm Axial CT with three contiguous slices taken two slices apart demonstrating significant involution of the nodule of interest, now measuring 4 mm. (c) (May 2024) 1 mm Axial CT with three contiguous slices taken two slices apart demonstrating a significant increase in the size of the pulmonary nodule to 15 mm.

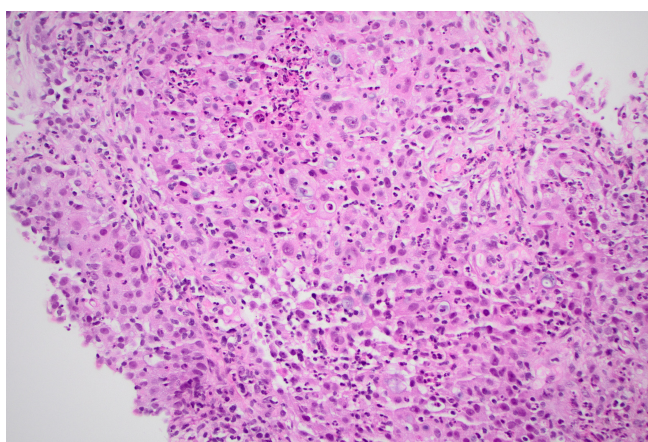


FIGURE 2 | The H&E sections demonstrate an adenocarcinoma with a predominantly solid architecture associated with scattered acute inflammatory cells. Intracytoplasmic mucin is focally present. Immunohistochemistry for TTF-1 and Napsin A was positive in the tumour (not shown).

consolidation within both lung bases. Accordingly, a follow-up CT chest and respiratory appointment to confirm the resolution of the consolidation was arranged.

CT chest completed in March 2022 demonstrated resolution of the previous consolidation; however, there was a new 8 mm solid subpleural nodule noted in the lateral basal segment of the right lower lobe (RLL) (Figure 1). As the patient was asymptomatic at the time of review, follow-up was arranged with a CT chest in June 2022, demonstrating stability and a CT chest in December 2022, demonstrating significant involution of the nodule, now measuring 4 mm (Figure 1b). At this point, the nodule was assumed to be likely benign, but given a high-risk smoking history, a further scan was arranged 1 year later, which demonstrated the nodule had increased slightly to 5 mm. Subsequently, the patient underwent a computed tomography pulmonary angiogram (CTPA) in May 2024 after developing post-operative hypoxia from a surgical procedure. On this scan, the RLL nodule dramatically increased in size to 15.5 mm (Figure 1c). A percutaneous CT-guided biopsy confirmed the diagnosis of

adenocarcinoma (Figure 2), and subsequent fluorodeoxyglucose positron emission tomography (FDG PET) confirmed localised disease (T1N0M0, Stage 1a). As the patient was not a surgical candidate due to the severity of his COPD, he underwent curative intent stereotactic radiotherapy.

3 | Discussion

Current guidelines do not address the management of nodules that have partially regressed in size. Case reports of spontaneous regression of lung cancer have been published in the literature, but remain rare, whereas reports of progression after initial spontaneous regression are rarer still. One case presented as a conference abstract describes a 9-mm solid nodule that slightly decreased in size at 3-month CT but eventually progressed and was diagnosed as small cell lung carcinoma [8]. Another case reports the progression of lymph node metastases despite complete radiological remission of the primary pulmonary adenocarcinoma post biopsy [9]. The authors postulated that forceps biopsy may have resulted in an immunological response in the primary tumour. In contrast to this case, our case describes spontaneous regression of adenocarcinoma without biopsy, followed by subsequent progression with no lymph node or distant metastases. Although it is possible that the initial nodule in our case was inflammatory and subsequently de novo adenocarcinoma developed, we believe this is unlikely, given only partial resolution of the initial nodule and subsequent imaging demonstrating the nodule in the exact same location after matching CT slices. The mechanism of partial resolution in our case remains unclear; however, it may be related to the initial pneumonia-provoking infection-mediated immune system activation.

Although rare, lung nodules that demonstrate partial spontaneous regression may still have malignant potential. As more nodules are detected as part of nationwide screening programmes, guidelines should consider following up partially resolved nodules to complete resolution.

Author Contributions

Lin Mo: conceptualisation, investigation, writing – original draft, writing – review and editing. **Filipe Vilas Boas:** visualisation, writing – review and editing. **Shileigh Martin:** visualisation, writing – review and editing. **Jessica Rajanayagam:** visualisation. **Vicki Pretorius:** visualisation, writing – review and editing.

Acknowledgements

Open access publishing facilitated by Flinders University, as part of the Wiley - Flinders University agreement via the Council of Australian University Librarians.

Consent

The authors declare that written informed consent was obtained for the publication of this manuscript and accompanying images using the consent form provided by the Journal.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data sharing is not applicable to this article as no datasets were generated or analysed during the current study.

References

1. J. Huang, Y. Deng, M. S. Tin, et al., “Distribution, Risk Factors, and Temporal Trends for Lung Cancer Incidence and Mortality: A Global Analysis,” *Chest* 161, no. 4 (2022): 1101–1111.
2. The National Lung Screening Trial Research Team, “Reduced Lung-Cancer Mortality With Low-Dose Computed Tomographic Screening,” *New England Journal of Medicine* 365, no. 5 (2011): 395–409.
3. H. J. Koning, C. M. Aalst, P. A. Jong, et al., “Reduced Lung-Cancer Mortality With Volume CT Screening in a Randomized Trial,” *New England Journal of Medicine* 382, no. 6 (2020): 503–513.
4. A. McWilliams, M. C. Tammemagi, J. R. Mayo, et al., “Probability of Cancer in Pulmonary Nodules Detected on First Screening CT,” *New England Journal of Medicine* 369, no. 10 (2013): 910–919.
5. M. E. J. Callister, D. R. Baldwin, A. R. Akram, et al., “British Thoracic Society Guidelines for the Investigation and Management of Pulmonary Nodules: Accredited by NICE,” *Thorax* 70, no. Suppl 2 (2015): ii1.
6. J. Christensen, A. E. Prosper, C. C. Wu, et al., “ACR Lung-RADS v2022: Assessment Categories and Management Recommendations,” *Journal of the American College of Radiology* 21, no. 3 (2024): 473–488.
7. H. MacMahon, D. P. Naidich, J. M. Goo, et al., “Guidelines for Management of Incidental Pulmonary Nodules Detected on CT Images: From the Fleischner Society 2017,” *Radiology* 284, no. 1 (2017): 228–243.
8. C. Parker and C. R. Sears, “Small Cell Lung Cancer Presenting as a Shrinking Nodule—A Cautionary Tale,” *American Journal of Respiratory and Critical Care Medicine* 211 (2025): A4919.
9. K. Kaira, H. Imai, A. Mouri, O. U. Yamaguchi, and H. Kagamu, “Progression to Lymph Node Metastasis After Spontaneous Regression of Pulmonary Adenocarcinoma Following Biopsy,” *In Vivo (Athens, Greece)* 38, no. 3 (2024): 1498–1502.